

What are the determinants of Brain Weight?

Christopher Rivera

Abstract:

Humans have evolved the largest brains per body mass in the animal kingdom. Large brains are essential to humans ability to adapt, socially interact, and plan. Decline in brain mass is associated with aging and cognitive decline. Thus understanding factors that are associated with brain mass are important. In this project, I attempt to understand how brain mass in humans is related to head size, sex and age. To do this I conduct analysis of a historical data set that includes brain masses, head size, gender and age group for 247 individuals. I first conduct exploratory data analysis followed by Bayesian linear regression analysis. I find that brain mass is positively correlated with head size (circumference) and reduce with female sex and older age.

Introduction

One of the great innovations afforded to humans is a large brain, which allows us to solve complex problems and interact socially. The human skull and head size has evolved to encapsulate our large brains. An interesting question is what determines head size?

To partially address this question, researchers have collected data for adults relating head size (volume in cm^3), gender (male or female) and age range (20-46 versus 46+) to brain weight (grams).

In this document, I present analysis of this data using exploratory data analysis and Bayesian linear regression.

Source: R.J. Gladstone (1905). “A Study of the Relations of the Brain to to the Size of the Head”, Biometrika, Vol. 4, pp105-123

Data

As noted above, the data consisted of 247 rows and 4 columns. The columns are Gender, a categorical with values Male = 1 and 2=Female, Age Group (also categorical) with values 1 if Age $\leq$ 45 and 2 if Age $>$ 45, Head size (cm^3) and Brain Mass (grams).

Initially, I checked for missing values (of which there were none). To aid with modeling, I converted the categorical variables into binary Indicator variables by subtracting 1. This resulted in 2 new variables which I used for exploratory analysis and modeling. These variables are female, 0 if male and 1 if female, older, 0 if age $\leq$ 45 and 1 otherwise. Finally, to make the data more amendable to modeling, I created a normalized version of the head size variable by centering and and scaling it.

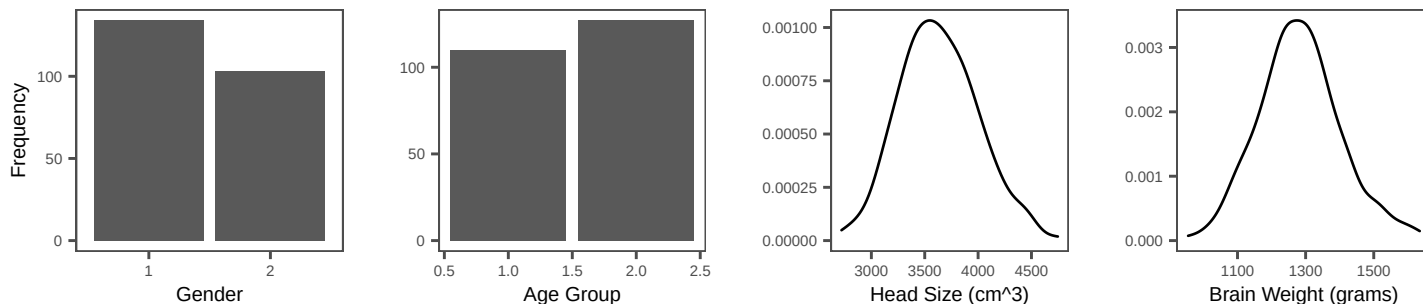
Table 1: First 5 rows of dataset

Gender	Age	Head	Weight	female	older	head_std
1	1	4512	1530	0	0	2.40
1	1	3738	1297	0	0	0.28
1	1	4261	1335	0	0	1.72
1	1	3777	1282	0	0	0.39
1	1	4177	1590	0	0	1.49
1	1	3585	1300	0	0	-0.13

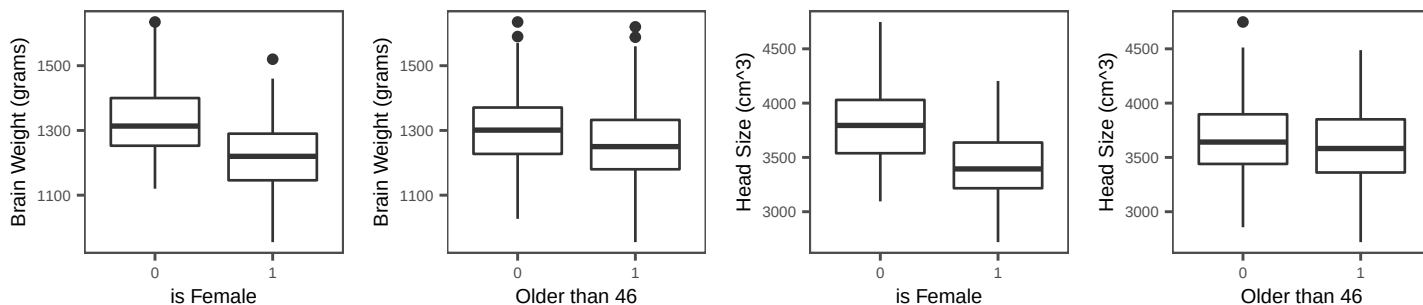
Initial exploratory analysis of the data. I conducted an initial exploratory data analysis to understand the data in more detail.

What’s the break down of gender and age?

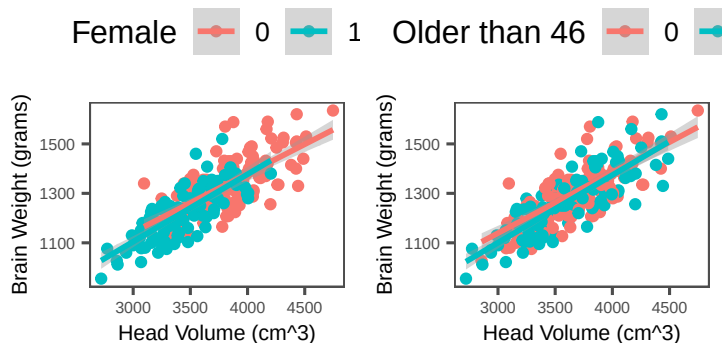
Below, I plot bar charts illustrating the number of individuals of each gender and age group. Note that although imbalances exist, they are not severe.



How does gender and age affect head size and brain weight? Below, I plotted box-plots to examine the relationship between the categorical variables and the continuous variables.



How does head volume affect brain weight? Below, I graph scatter plots of brain weight versus head volume. To examine the effects of gender and age, I color coded the points based on gender or age group. Note that we see a positive correlation between head volume and brain weight. Both gender and age do not really appear to affect the relationship (slope).



Models

To determine the relation between brain weight and the other variables in this data set, I fit several models of increasing complexity. For each model, I simulated 3 markov chains and initially ran 5000 iterations for burn-in. I ran 50000 iterations to collect parameter statistics. I visually inspected the resulting markov chains and looked at the auto-correlations to assess the quality of the MC dynamics.

Model 1 assumes a simple linear relationship between brain weight and head size, where h_i is standardized.

$$\begin{aligned}
w_i &\sim \mathcal{N}(\mu_i, 1/\sigma^2) \\
\mu_i &= b_0 + b_1 h_i \\
b_0 &\sim \mathcal{N}(0, 10^{-6}) \\
b_1 &\sim \mathcal{N}(0, 10^{-6}) \\
\sigma^2 &\sim \text{IG}(2, 1)
\end{aligned}$$

Model 2 adds add the age indicator variable.

$$\begin{aligned}
w_i &\sim \mathcal{N}(\mu_i, 1/\sigma^2) \\
\mu_i &= b_0 + b_1 h_i + b_2 I_{\text{age}>45} \\
b_0 &\sim \mathcal{N}(0, 10^{-6}) \\
b_1 &\sim \mathcal{N}(0, 10^{-6}) \\
b_2 &\sim \mathcal{N}(0, 10^{-6}) \\
\sigma^2 &\sim \text{IG}(2, 1)
\end{aligned}$$

Model 3 includes gender indicator.

$$\begin{aligned}
w_i &\sim \mathcal{N}(\mu_i, 1/\sigma^2) \\
\mu_i &= b_0 + b_1 h_i + b_2 I_{\text{female}=\text{True}} \\
b_0 &\sim \mathcal{N}(0, 10^{-6}) \\
b_1 &\sim \mathcal{N}(0, 10^{-6}) \\
b_2 &\sim \mathcal{N}(0, 10^{-6}) \\
\sigma^2 &\sim \text{IG}(2, 1)
\end{aligned}$$

Model 4 used all three variables as predictors.

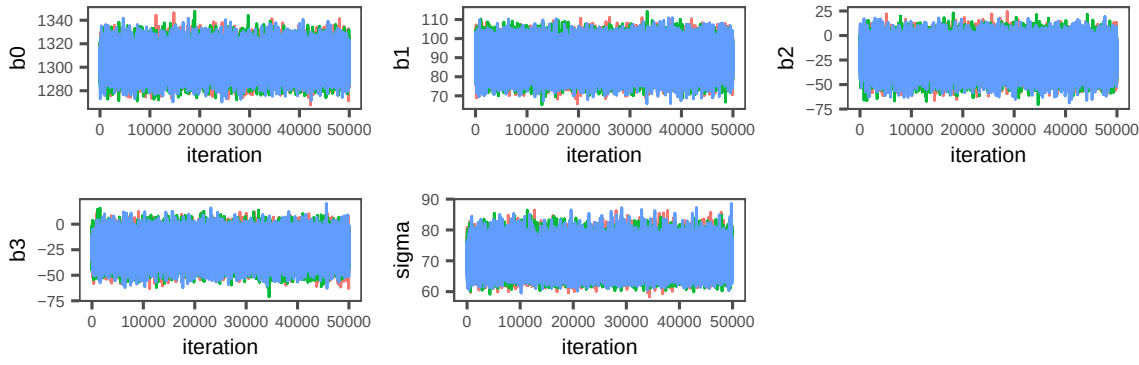
$$\begin{aligned}
w_i &\sim \mathcal{N}(\mu_i, 1/\sigma^2) \\
\mu_i &= b_0 + b_1 h_i + b_2 I_{\text{female}=\text{True}} + b_3 I_{\text{age}>45} \\
b_0 &\sim \mathcal{N}(0, 10^{-6}) \\
b_1 &\sim \mathcal{N}(0, 10^{-6}) \\
b_2 &\sim \mathcal{N}(0, 10^{-6}) \\
b_3 &\sim \mathcal{N}(0, 10^{-6}) \\
\sigma^2 &\sim \text{IG}(2, 1)
\end{aligned}$$

Results

I selected the best model based upon DIC. The table below shows the DIC results.

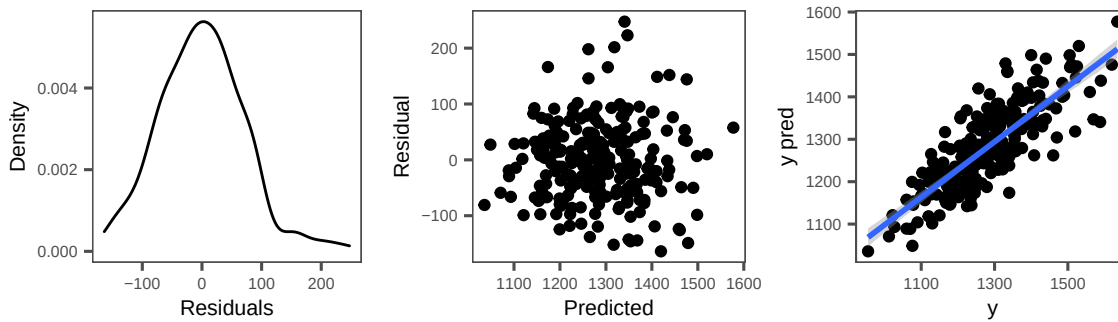
	Deviance	penalty	Penalized Deviance
Model 1	2704	2.995	2707
Model 2	2700	4.069	2704
Model 3	2702	4.108	2706
Model 4	2696	4.859	2701

As can be seen the best model (model 4) employs all 3 predictors. Below are plots for model 4's markov chains. Note, that the the chains are well mixed.



To further understand the goodness of fit, I again plot a KDE of the residuals, the predicted y values versus the residuals and the predicted values of brain weight versus the true values. As we can see this model again has departure of the residuals way from a normal distribution and some heteroskedasticity .

Residual plots for Model 4

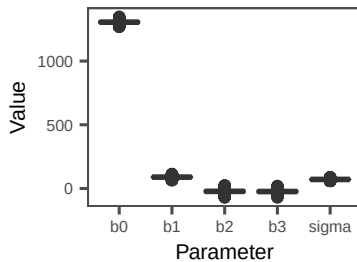


Finally, I estimate the mean parameter values.

	mean	SE
b0	1305.47	0.02
b1	89.20	0.01
b2	-22.51	0.03
b3	-23.97	0.02
sigma	70.98	0.01

As suggested by our exploratory analysis, both female gender and older than 46 are associated with a smaller brain size.

Boxplots of the distributions of the parameters indicate that these are good estimates given the data are are definitely non zero.



Conclusions:

In this document, I explored the relationship between brain weight and head size (cm^3), gender and age. I fit 4 models and compared them based on DIC. The best model as indicated by lowest DIC was of the form $y_i = 1305.5 + 89.2h_i - 22.5I_{\text{female}=\text{True}} - 24.0I_{\text{age}>46} + \epsilon_i$. While it violates the assumptions of linear regression (from a frequentist point of view), it is a decent model.